

Cross-Domain Threat Detection: XAI and Ensemble ML for IIoT and Darknet Scenarios

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Abstract—In the age of ever-evolving cyber-attacks, IoT, and darknet, security needs new and understandable methods. This paper presents a cross-domain investigation of intrusion detection on the CIC-APT-IIoT-2024 and CIC-Darknet2020 datasets with the performance of high-performance machine learning algorithms and an ensemble. It also investigates the application of XAI (Explainable AI) approaches like LIME and SHAP to bring more model visibility and helpful information about cross-domain threat detection. Using random forest models, XGBoost, RNN-LSTM, Transformers, and ensembles such as Stacking and Voting Classifiers, the work gets brilliant results. Ensemble models achieved near-perfect 99% accuracy on CIC-APT-IIoT-2024 and 98% accuracy on CIC-Darknet2020, while precision, recall, and F1-scores were equally good, underscoring their potential for scalable and reliable cross-domain security solutions.

Index Terms—Cross-Domain Security, CIC-APT-IIoT-2024, CIC-Darknet2020, Ensemble ML, Explainable Artificial Intelligence (XAI), Intrusion Detection Systems

I. INTRODUCTION

In this era of evolving cyber-attacks and sophisticated threats, IoT networks and darknet security have been a major concern [1] [2]. Traditional security mechanisms generally fail to detect cross-domain threats [3] [4], and thus critical systems remain exposed to attacks that exploit both IoT environments and darknet traffic. This requirement is quite timely because the rapidly growing sophistication of cyber threats and increasingly connected devices places an excellent need for sound intrusion detection systems that will perform well across these domains.

The increasing adoption of IoT devices and reliance on darknet communication channels introduce new security challenges [5]. While existing intrusion detection models perform well in single-domain applications, cross-domain threat detection remains underexplored. This research addresses this gap by applying advanced machine learning models to the CIC-APT-IIoT-2024 and CIC-Darknet2020 datasets, focusing on scalable solutions for detecting threats across diverse network environments.

Our experiment demonstrates that machine learning models are very effective for cross-domain threat detection. SVM and Logistic Regression obtained 99% accuracy on the CIC-APT-

IIoT-2024 dataset, and XGBoost and Stack Ensemble of XGB, LR, and RF obtained 98% on CIC-Darknet2020. These results demonstrate the power of conventional and ensemble ML methods in improving IoT network security and darknet threat detection. Applying XAI techniques such as LIME and SHAP further enhances transparency and interpretability, allowing for increased trust among cybersecurity professionals and more efficient threat mitigation.

To our knowledge, this is the first in-depth analysis of the CIC-APT-IIoT-2024 dataset for intrusion detection. Previous works have focused on the CIC-Darknet-2020 dataset, but no study has examined the cross-domain applicability between IIoT and darknet environments. Our work fills this research gap by providing an in-depth evaluation of both datasets and showing the feasibility of cross-domain threat detection using different machine-learning models.

The paper is structured as follows: Section II reviews related works on intrusion detection and cross-domain security, while Section III details the datasets, preprocessing, and ML methodologies. Sections IV, and V, cover implementation and results, and conclusion emphasizing key findings, contributions, and recommendations for advancing cross-domain intrusion detection.

II. LITERATURE REVIEW

The following section reviews the existing literature on IoT network security and Darknet security, focusing on applying ensemble methods and Explainable AI (XAI) techniques.

A. Advancements in CIC-Darknet2020 dataset studies

The studies below specifically highlight the existing advancements in the CIC-Darknet2020 dataset.

In their study, Pookpun *et al.* [6] classified darknet traffic using the CIC-Darknet2020 dataset with eight user behavior classes. Models were trained using random search, including SMOTE, ADASYN, Random Forest (with XGBoost), and LightGBM. Feature selection was based on mutual information, with normalization via Min-Max scaling. Random Forest with SMOTE achieved the highest accuracy at 93.1%. However, the study's focus was limited to two oversampling methods, highlighting the need for exploring additional data

balancing and feature selection techniques to improve performance.

Mohsin *et al.*'s [7] research study describes an Ensemble Method with Meta-Learning for DarkNet Traffic Recognition Stacking on Random Forest, XGBoost, Catboost, and Adaboost for the CIC-Darknet2020 dataset. It uses preprocessing for missing values, feature selection, and normalization. Stacking the ensemble (98.8% accuracy) was better than single models.

In the study by Yang *et al.* [8], a new autoencoder-based darknet traffic identification algorithm using an improved feature selection algorithm (AE-FS) and a CNN model is proposed. AE-FS maps features from the data into greyscale images, assigns ranking according to reconstruction error, and saves just key features for dimension reduction. Dropout layers on the CNN detect traffic and limit overfitting. Based on experimental proofs using datasets such as ISCXTor2016 and CIC-Darknet2020 AE-DTI, it is more accurate, general, and efficient than PCA or LLE.

In their article, Kirubavathi *et al.* [9] proposed an attention-LSTM model for darknet traffic detection and analysis, supported by XAI methods like LIME. Preprocessing the CIC-Darknet2020 dataset involved duplicate detection, SMOTE balancing for class distribution, and feature encoding. The model achieved 99.7% detection accuracy and 97% characterization, outperforming current methods.

The datasets similar to the ones used in this study, CIC-IoV-2024 [10] and CIC-IoMT-2024 [11], provide benchmark resources for advancing cybersecurity in IoV and IoMT domains. CICIoV2024 focuses on spoofing and DoS attacks executed on a 2019 Ford car's ECUs, enhancing machine learning-based IoV security, while CICIoMT2024 features 18 diverse attacks across multi-protocol healthcare devices to strengthen security mechanisms in medical systems.

B. Ensemble Methods and Explainable AI

The following studies discuss applying ensemble methods and Explainable AI (XAI) techniques in IoT scenarios and darknet traffic analysis on other datasets and scenarios.

The study by Gaspar *et al.* [12] considers using XAI technology, LIME, and SHAP to describe black-box machine learning algorithms in IoT intrusion detection systems. This results in a more easily understandable model and cybersecurity decision-making. However, there are limits, such as the computational burden, especially of SHAP, and the need to expand the XAI methods and datasets to get more generalizability and robustness.

The paper by Shi *et al.* [13] Intrusion detection ensemble system for IoT security using Enhanced Artificial Hummingbird Algorithm (EAHA) to optimize feature selection and multiple machine learning classifiers for detection. EAHA optimizes on the original algorithm with hunger-driven foraging, cooperative algorithms, and Cauchy mutation — run on benchmark functions. On NSL-KDD, CIC-IDS2017, and CSE-CIC-IDS2018 datasets, the feature reduction (up to 69%)

and accuracy (98.51%-99.74%) are shown in experimental experiments.

C. Research Gap

This study combines ensemble methods and XAI algorithms (LIME and SHAP) to comprehensively compare the CIC-APT-IIoT-2024 and CIC-Darknet2020 datasets, overcoming the limitations of prior work focused on specific models or datasets. Our ensemble techniques—stacking and voting classifiers—enhance flexibility and efficiency against IoT and darknet threats, while LIME and SHAP reveal key predictive features in model decisions. Additionally, we analyze model size and parameter count.

III. METHODOLOGY

In this section, we systematically compare intrusion detection models on two sets of data, CIC-APT-IIoT-2024 and CIC-Darknet2020. These datasets comply with best practices so that the data quality, relevance, and usability for a robust performance analysis are not compromised.

The CIC-APT-IIoT-2024 dataset employs binary classification, labeling regular traffic as zero and attack traffic as one, enabling efficient training of intrusion detection systems to differentiate benign from malicious activity. Similarly, the CIC-Darknet2020 dataset uses binary classification to distinguish between secured and unsecured network traffic. Secured traffic, including Tor and VPN types (e.g., Tor_Browsing, VPN_Email, VPN_Video-Streaming), is labeled as 0, while unsecured traffic, such as Non-Tor and NonVPN categories (e.g., NonVPN_File-Transfer, Non-Tor_Audio-Streaming), is labeled as 1. This labeling scheme facilitates the training of machine learning models to predict secure or insecure traffic, aiding in risk identification and mitigation.

A. Hardware Used

This study utilized cloud-based T4 GPUs.

B. Datasets

A.1 The CIC-APT-IIoT-2024 Dataset

The CIC-APT-IIoT-2024 dataset [14] is designed for cybersecurity research, focusing on detecting Advanced Persistent Threats (APTs) in Industrial IoT environments. Built using a hybrid testbed of real and hypothetical IIoT systems, it reflects the complexity of modern IIoT setups. The dataset simulates an APT campaign inspired by the APT29 attack group, featuring over 20 attack strategies classified into eight principal categories. It includes provenance logs (process-artifact relationship graphs) and network logs (traffic data recorded with NS3). Provenance data (CSV) contains metadata-rich nodes and edges, while network logs (pcap and CSV) capture network activity. Companion resources include scripts for network traffic processing, provenance graph generation, and node embedding creation. The dataset serves as a robust baseline for ML models in APT detection and supports cross-domain intrusion detection research.

A.2 The CIC-Darknet2020 Dataset

The CIC-Darknet2020 dataset [15] is a comprehensive darknet traffic dataset that combines ISCXTor2016 and IS-CXVPN2016 datasets to model VPN/TOR traffic by category (Audio-Stream, Browsing, Chat, Email, P2P, Transfer, Video-Stream, VOIP). It is bi-layered, where the top level is for benign traffic, and the bottom level has applications such as a darknet traffic generator. This is an excellent dataset for finding and segmenting darknet activity for detection and classification in real-time so that malware and malicious activity can be detected early.

C. Data Pre-Processing and Feature Engineering

The data preprocessing pipeline begins with data cleaning and transformation, including handling errors, removing duplicates, and ensuring feature consistency through data type conversions. Outliers are identified and removed to prevent negative impacts on the model. Features are scaled and standardized using StandardScaler to maintain consistent ranges and distributions. The effect of scaling on model precision and stability is also evaluated.

Feature engineering involves a structured, iterative process to refine datasets for machine learning. Random Forest feature importance and LDA plots identify meaningful features, while irrelevant columns (e.g., IDs) are removed. Interaction features are created to capture complex relationships, and less significant features are discarded or combined to reduce redundancy. Techniques like log scaling and binning enhance feature representation. The final feature set (X) and target variable (y) are prepared, and PCA is applied to reduce dimensionality while preserving variance. The data is then split into training and testing subsets (80-20 split) using stratified sampling, with additional validation sets created for robust evaluation.

D. Models

For all the models, including Machine Learning, Deep Learning, Ensemble, and Ensemble models, we used Explainable AI methods like LIME and SHAP for transparent and interpretable models. Such approaches gave features importance and model selection information needed to fully understand how predictions were made on CIC-APT-IIoT-2024 and CIC-Darknet2020 datasets.

For a comprehensive overview, please refer to Table. I, which summarizes each model's key parameters and objectives.

C.1 Machine Learning Models

The machine learning intrusion detection model was applied to the CIC-APT-IIoT-2024 and CIC-Darknet2020 datasets with preprocessing steps including handling missing values, encoding categorical columns (Source IP, Destination IP, Protocol Name), and selecting numerical features. Algorithms like Random Forest (100 estimators), SVM (RBF kernel), XGBoost (100 estimators, 0.1 learning rate), AdaBoost (Decision Tree base estimator with max depth 1), Logistic Regression (1,000

max iterations), and KNN (k=5) were deployed with hyperparameter tuning. Feature scaling was used for SVM, KNN, and Logistic Regression, and an 80-20 stratified train-test split ensured class diversity. Each model underwent 5-fold cross-validation, and performance was evaluated using accuracy, precision, recall, and F1-score. Adjustments to learning rates and KNN k-values further optimized model performance, enabling reliable and effective results for both datasets.

C.2 Deep Learning Models

Deep learning models used across both datasets included RNN+LSTM, MLP, Transformers, Autoencoders, and CNNs. Preprocessing involved handling missing values, encoding categorical columns (Source IP, Destination IP, Protocol Name), and standardizing numerical features using StandardScaler. RNN+LSTM combined SimpleRNN and LSTM with ReLU and sigmoid activation, trained on 64 batches for 10 epochs with binary cross-entropy loss. MLP utilized dense layers with dropout regularization, trained over 10 epochs with 64 batches using the Adam optimizer. Transformers incorporated multi-head attention, deep layers, and global average pooling, trained for five epochs with a 1e-4 learning rate. Autoencoders focused on anomaly detection via feature reconstruction, trained for 50 epochs with 32 encoding sizes. CNNs employed Conv1D and GlobalMaxPooling, trained for 50 epochs on 32 batches with categorical cross-entropy loss. An 80-20 train-test split and 5-fold cross-validation ensured robustness, with hyperparameters like learning rates, batch sizes, layers, and neurons tuned for optimal performance. Models were evaluated on accuracy, precision, recall, and F1 score.

C.3 Ensemble Methods

For the datasets, ensemble learning with stacking and voting classifiers was adopted. The stacking classifier was a combination of XGBoost, Random Forest, and Logistic Regression, where XGBoost used a default learning rate of 0.1 and log-loss as the measure, Random Forest used default parameters and random state 42, and Logistic Regression was the meta-classifier. Soft Voting Classifier using XGBoost and Logistic Regression. They used stratified splits of train-test (80-20) to have equally distributed classes. Models were compared by accuracy, precision, recall, and F1 scores.

In this research study, we applied the methods above to the CIC-APT-IIoT-2024 and CIC-Darknet2020 datasets to identify the optimal model based on its performance, size, and parameter count.

IV. IMPLEMENTATION AND ANALYSIS

Refer to Table. I for the summary of the implementation of the models included in the research study. Implementing machine-learning or deep-learning algorithms starts with training single algorithms for a specific purpose. Random Forest (RF) is trained with 100 estimators for underfitting and accuracy. Meanwhile, gradient boosting and weak learner boosting are employed by XGBoost and AdaBoost, with 100 estimators trained at rates appropriate to their purpose. Logistic

TABLE I
OVERVIEW OF MACHINE LEARNING, DEEP LEARNING, AND ENSEMBLE MODELS

Model	Key Parameters	Objective
Random Forest (RF)	100 estimators, state 42, n-jobs = 1	Reduce overfitting and improve accuracy
XGBoost	LR = 0.1, 100 estimators, max depth = 6	Optimize gradient boosting
AdaBoost	Decision tree base estimator, 100 estimators, LR = 0.1	Boost weak learners
Logistic Regression (LR)	Balanced weights, max_iter = 1000	Address class imbalance
K-Nearest Neighbors (KNN)	5 neighbors, Minkowski metric	Classify based on proximity
CNN	1 Conv1D layer, global max pooling, 1 dense layer	Capture spatial hierarchies
MLP	2 hidden layers, ReLU, dropout 0.3, 10 epochs	Model non-linearity
RNN + LSTM	1 RNN layer, 1 LSTM layer, dropout 0.5, 10 epochs	Capture temporal dependencies
Transformers	4 transformer blocks, head size = 256, LR = 1e-4	Model complex relationships
Autoencoders	Encoding layer (32 units), LR = 0.001, 50 epochs	Detect anomalies using reconstruction errors
Stacking Ensemble	Combine RF, XGBoost, Logistic Regression	Enhance accuracy through aggregation
Voting Ensemble	Soft voting with RF, XGBoost, NN	Aggregate predictions for better accuracy

Regression (LR) with class weighted balanced to correct for class bias, and K-Nearest Neighbors (KNN) Minkowski with five neighbors used for proximity classification. The deep learning architectures are a CNN to learn spatial hierarchy (Conv1D layer, global max pooling), a Multi-Layer Perceptron (MLP) with ReLU activation and dropout for non-linearity, and an RNN-LSTM for temporal dependence (recurrent and LSTM layers with dropout regularization).

Stacking, voting, and so on are used in ensembles to improve model performance. The stacking ensemble merges RF, XGBoost, and LR predictions with model variation for better performance. Voting ensemble: soft voting combined with RF, XGBoost, and neural networks for collective prediction. There are also transformers for the complex relationship representation with multiple transformer blocks and autoencoders for anomaly detection based on reconstruction error. Based on standardized datasets, such models have tailored hyperparameters for maximum performance for each task while keeping adaptability and robustness in the overall system.

A. Stage-I: Using the CIC-Darknet2020 dataset

Table II summarizes the results of applying various models to the CIC-Darknet2020 dataset.

The experiment on the CIC-Darknet2020 dataset shows that traditional ML algorithms like Random Forest, Decision Tree, and XGBoost perform well with F1 scores and accuracies of 0.99. Logistic Regression, with an F1 Score of 0.96 and a compact size of 7.25 KB, is ideal for resource-constrained environments. KNN achieves a high F1 Score of 0.99 but is inefficient due to its 400,000 parameters, making it computationally expensive to deploy.

Deep learning models effectively handle complex patterns, with RNN-LSTM achieving an F1-score of 0.97 using 25,391 parameters, while MLP slightly outperforms it with an F1-score of 0.98 and moderate model size. CNNs excel in spatial feature analysis with an F1-score of 0.92, whereas Transformers and Autoencoders struggle, with Autoencoders proving unsuitable for classification (F1-score: 0.08). Ensemble models demonstrate versatility, with a stacking ensemble (Random Forest, Logistic Regression, XGBoost) achieving an F1-score of 0.99 on CIC-Darknet2020, maintaining high precision, recall, and efficiency (24.9 KB, 620,561 parameters). The Voting

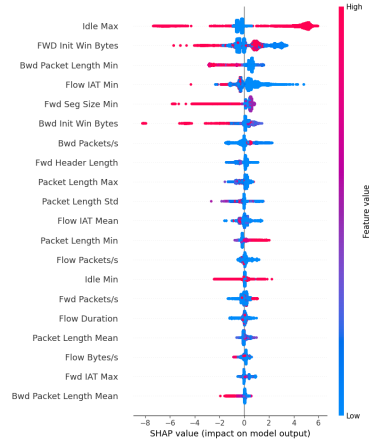


Fig. 1. Interpreting Ensemble techniques Stack Ensemble of XGB, LR, RF Predictions Using Shap Visualization on The CIC-Darknet2020 dataset

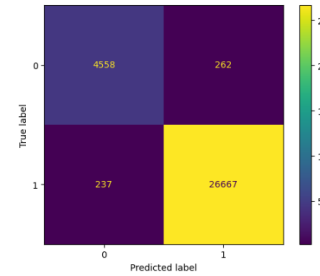


Fig. 2. Confusion Matrix of Ensemble techniques Stack Ensemble of XGB, LR, RF

Classifier also delivers strong performance with a smaller size (16.8 KB), reinforcing the reliability of ensemble methods. Logistic Regression remains a lightweight yet practical choice for resource-constrained environments.

The LIME visualization from the study to analyze the CIC-Darknet202 dataset shows that, among others, the 'Fwd_PSH_Flags' and 'Fwd_Seg_Size_Min' features are 98% confident of predicting a "Malicious" instance. Figure. 1 SHAP summary plot identifies global feature importance, where 'Idle Max', 'Fwd Init Win Bytes', and 'Bwd Packet Length Min' are the most impactful. Both methods suggest that the stack

TABLE II
PERFORMANCE METRICS AND MODEL PARAMETERS COMPARISON ACROSS DIFFERENT MODELS USING THE CIC-DARKNET2020 DATASET

Model	Performance Metrics				Model Parameters and Hyperparameters		
	Precision	Recall	F1-Score	Accuracy	Model Size (KB)	No. of Parameters	No. of Hyper-Parameters
Random Forest	0.99	0.99	0.99	0.98	40.25	4850	17
Decision Tree	0.98	0.98	0.98	0.97	24.70	50	5
XGBoost	0.98	0.99	0.99	0.98	77.80	3826	12
Logistic Regression	0.95	0.97	0.96	0.93	7.25	51	12
Adaboost	0.96	0.97	0.96	0.94	328	290	5
KNN	0.98	0.99	0.99	0.98	3200	400,000	5
MLP	0.97	0.98	0.98	0.96	58.3	14,914	12
RNN+LSTM	0.98	0.97	0.97	0.96	329	25,391	14
Transformers	0.93	0.99	0.96	0.93	73.78	4,134,696	14
CNN	0.85	0.99	0.92	0.85	8.91	6910	12
Auto Encoders	0.76	0.04	0.08	0.18	43.11	12,373	12
Stack Ensemble (RF, Decision Tree, GCLR)	0.99	0.99	0.99	0.98	24.9	620,561	13
Stack Ensemble (XGB, LR, RF)	0.99	0.99	0.99	0.98	25.2	620,079	12
Voting Classifier (XGB, LR, RF - Soft Voting)	0.99	0.97	0.98	0.97	16.8	460,000	11

ensemble model was highly accurate in classifying 4,557 "Not Malicious" instances out of 4,821 and 26,668 "Malicious" instances with over 98% classification accuracy.

The confusion matrix in Fig. 2 shows excellent classification, as it correctly predicted 4,557 out of 4,821 "Not Malicious" and 26,668 "Malicious" instances, with very few false positives and false negatives, indicating more than 98% accuracy.

B. Stage-2: Using the CIC-APT-IIoT-2024 dataset

Table. III summarizes the results of applying various models to the CIC-APT-IIoT-2024 dataset.

The evaluation on the CIC-APT-IIoT-2024 dataset shows strong performance, with Random Forest, SVM, and Logistic Regression achieving 0.99 in precision, recall, F1-score, and accuracy. Random Forest balances performance and complexity (4,860 parameters, 18 hyperparameters), SVM offers simplicity (16 parameters, six hyperparameters), and Logistic Regression is lightweight and efficient (7.75 KB, 51 parameters). KNN scored 0.97 but is computationally expensive with 400,000 parameters. Deep learning models like RNN-LSTM achieved 0.99 (25,401 parameters, 15 hyperparameters) but with a larger size (339 KB). CNNs scored 0.89 (9.01 KB), MLP achieved 0.98 (14,849 parameters), while Transformers, despite their complexity (4,134,706 parameters), underperformed with 0.76 accuracy. Autoencoders, designed for anomaly detection, struggled with 0.73 precision and recall, showing limited suitability for classification tasks.

Ensemble methods excelled in performance, with the stacking ensemble (Random Forest, SVM, Logistic Regression) achieving 0.99 across all metrics, demonstrating strong synergy but requiring high computational resources (630,102 parameters). The voting classifier (Random Forest, XGBoost with soft voting) also reached 0.99, offering a balance of accuracy and efficiency with 470,000 parameters and a smaller model size (17.9 KB). On the CIC-APT-IIoT-2024 dataset, Random Forest, SVM, XGBoost, Logistic Regression, RNN-LSTM, and ensemble models achieved 0.99 in precision, recall, F1-score, and accuracy. Logistic Regression was the most efficient (7.75 KB, 51 parameters), making it ideal for resource-constrained scenarios. While ensemble models provided high accuracy through diversity, Transformers, and

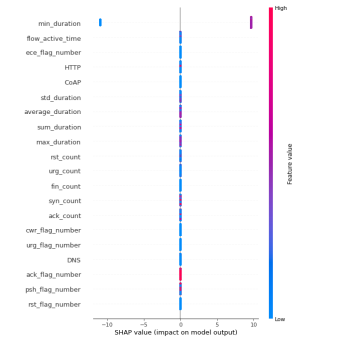


Fig. 3. Interpreting XGBoost Predictions Using SHAP Visualization on The CIC-APT-IIoT-2024 dataset

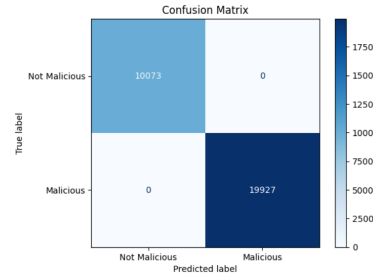


Fig. 4. Confusion Matrix for XGBoost Model Evaluation

Autoencoders underperformed, proving unsuitable for intrusion detection in IoT environments.

The illustration below showcases the best-performing algorithm, XGBoost, on the CIC-APT-IIOT-2024 dataset.

In the research study, we used LIME to demonstrate how individual predictions by the XGBoost model on the CIC-APT-IIoT-2024 dataset were given in LIME. It highlights feature contributions (e.g., min_duration and UDP) to the "Malicious" vs. "Not Malicious" classification decision. The second image, Fig. 3, uses SHAP to describe the XGBoost model globally. The SHAP summary plot shows the whole influence of features on the prediction made by the model, with min_duration and flow_active_time being the most important. These visualizations provide better readability as they present local and global feature weights.

The confusion matrix in Fig. 4 is performing perfectly

TABLE III
PERFORMANCE METRICS AND MODEL PARAMETERS COMPARISON ACROSS DIFFERENT MODELS USING THE CIC-APT-IIoT-2024 DATASET

Model	Performance Metrics				Model Parameters and Hyperparameters		
	Precision	Recall	F1-Score	Accuracy	Model Size (KB)	No. of Parameters	No. of Hyper-Parameters
Random Forest	0.99	0.99	0.99	0.99	41.82	4,860	18
SVM	0.99	0.99	0.99	0.99	25.90	16	6
XGBoost	0.99	0.99	0.99	0.99	78.11	3,874	13
AdaBoost	0.98	0.995	0.99	0.98	332	300	5
Logistic Regression	0.99	0.99	0.99	0.99	7.75	51	8
KNN	0.97	0.97	0.97	0.97	3,212	400,000	8
RNN + LSTM	0.99	0.99	0.99	0.99	339	25,401	15
MLP	0.98	0.98	0.98	0.98	59.4	14,849	13
Transformers	0.80	0.86	0.83	0.76	73.88	4,134,706	15
Auto Encoder	0.73	0.73	0.73	0.75	42.8	12,383	13
CNN	0.89	0.89	0.89	0.89	9.01	6,920	13
Stacking Ensemble (RF+SVM+LR)	0.99	0.99	0.99	0.99	25.2	630,102	15
Voting Classifier (RF+XGBoost)	0.99	0.99	0.99	0.99	17.9	470,000	12

for classification (10073 out of 19,927 "Not Malicious" vs. 19,927 "Malicious" prediction), without false positives or false negatives. This indicates 99% accuracy.

This study presents a novel cross-domain intrusion detection framework using Explainable AI and ensemble machine learning on the CIC-APT-IIoT-2024 and CIC-Darknet2020 datasets. Using XAI methods such as LIME and SHAP, the model achieves an accuracy of 99% and 98% by stacking and voting ensemble classifiers. This scalable and interpretable approach improves security in IIoT and darknet environments, which is a significant gap in the research of cross-domain cybersecurity.

V. CONCLUSION

CIC-APT-IIoT-2024 and CIC-Darknet2020 datasets. ML models, such as Random Forest and XGBoost, achieved over 99% accuracy on structured data, whereas DL models, especially RNN-LSTM, were good at handling complex threats. Stacking ensembles, namely RF+SVM+LR for CIC-APT-IIoT-2024, and RF+DT+GCLR for CIC-Darknet2020, achieved over 99% precision, recall, and F1-score, demonstrating their performance in malware detection. ML is well-suited for structured tasks, but DL is suitable for evasive threats, with XAI providing better interpretability. However, generalization across the unseen threat is still a challenge. Future work focuses on real-time deployment, adaptive learning, and testing in diverse cybersecurity environments.

However, limitations include challenges in generalizing across unseen threats, high computational demands for deep learning models, and the need for real-time deployment adaptability. Future research should focus on optimizing lightweight models for resource-constrained environments and enhancing adaptive learning mechanisms for evolving cybersecurity threats.

Data Availability: The datasets are publicly available, CIC-APT-IIoT-2024 dataset [14], and CIC-Darknet2020 dataset [15]. **Code Implementation:** <https://github.com/PapineniSaisharan/Cross-Domain-Threat-Detection-XAI-and-Ensemble-ML-for-IIoT-and-Darknet-Scenarios>

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